

THE TWENTY-SEVENTH WILLIAM LOWELL PUTNAM  
MATHEMATICAL COMPETITION

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A-1. Let  $f(n)$  be the sum of the first  $n$  terms of the sequence  $0, 1, 1, 2, 2, 3, 3, 4, \dots$ , where the  $n$ th term is given by

$$a_n = \begin{cases} n/2 & \text{if } n \text{ is even,} \\ (n-1)/2 & \text{if } n \text{ is odd.} \end{cases}$$

Show that if  $x$  and  $y$  are positive integers and  $x > y$  then  $xy = f(x+y) - f(x-y)$ .

A-2. Let  $a, b, c$  be the lengths of the sides of a triangle, let  $p = (a+b+c)/2$ , and  $r$  be the radius of the inscribed circle. Show that

$$\frac{1}{(p-a)^2} + \frac{1}{(p-b)^2} + \frac{1}{(p-c)^2} \geq \frac{1}{r^2}.$$

A-3. Let  $0 < x_1 < 1$  and  $x_{n+1} = x_n(1-x_n)$ ,  $n = 1, 2, 3, \dots$ . Show that

$$\lim_{n \rightarrow \infty} nx_n = 1.$$

A-4. Prove that after deleting the perfect squares from the list of positive integers the number we find in the  $n$ th position is equal to  $n + \{\sqrt{n}\}$ , where  $\{\sqrt{n}\}$  denotes the integer closest to  $\sqrt{n}$ .

A-5. Let  $C$  denote the family of continuous functions on the real axis. Let  $T$  be a mapping of  $C$  into  $C$  which has the following properties:

- 1.  $T$  is linear, i.e.  $T(c_1\psi_1 + c_2\psi_2) = c_1T\psi_1 + c_2T\psi_2$ , for  $c_1$  and  $c_2$  real and  $\psi_1$  and  $\psi_2$  in  $C$ .
- 2.  $T$  is local, i.e. if  $\psi_1 \equiv \psi_2$  in some interval  $I$  then also  $T\psi_1 \equiv T\psi_2$  holds in  $I$ .

Show that  $T$  must necessarily be of the form  $T\psi(x) = f(x)\psi(x)$  where  $f(x)$  is a suitable continuous function.

A-6. Justify the statement that

$$3 = \sqrt{1 + 2\sqrt{1 + 3\sqrt{1 + 4\sqrt{1 + 5\sqrt{1 + \dots}}}}}$$

B-1. Let a convex polygon  $P$  be contained in a square of side one. Show that the sum of the squares of the sides of  $P$  is less than or equal to 4.

B-2. Prove that among any ten consecutive integers at least one is relatively prime to each of the others.

B-3. Show that if the series

$$\sum_{n=1}^{\infty} \frac{1}{p_n}$$

is convergent, where  $p_1, p_2, p_3, \dots, p_n, \dots$  are positive real numbers, then the series

$$\sum_{n=1}^{\infty} \frac{n^2}{(p_1 + p_2 + \cdots + p_n)^2} p_n$$

is also convergent.

- B-4. Let  $0 < a_1 < a_2 < \cdots < a_{mn+1}$  be  $mn+1$  integers. Prove that you can select either  $m+1$  of them no one of which divides any other, or  $n+1$  of them each dividing the following one.
- B-5. Given  $n (\geq 3)$  distinct points in the plane, no three of which are on the same straight line, prove that there exists a simple closed polygon with these points as vertices.
- B-6. Show that all solutions of the differential equation  $y'' + e^x y = 0$  remain bounded as  $x \rightarrow \infty$ .